

MTU_ValueService **Technical Documentation**

Frac Pack with
MTU Series 4000 12 V 4000 T94L
PFRPK12V4000-1A0
Oil & Gas

Operation and Maintenance Manual
SS180036/01E



System designation	Engine model	kW/cyl.	Application group
PFRPK12V4000-1A0	12V4000T94L	155,4 kW/ cyl.	4D, Frac-operation (continuous operation with low load)

Table 1: Applicability

Printed in Germany

© 2013 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

1	Safety	7
1.1	General conditions	7
1.2	Personnel and organizational requirements	9
1.3	Safety Requirements when Working on the Engine-Transmission Package	11
1.3.1	Safety requirements in general	11
1.3.2	Safety while putting into or out of operation and storing	15
1.3.3	Safety during repair and maintenance	19
1.3.4	Special safety requirements for welding work	26
1.4	Auxiliary materials, fire prevention, noise control and environmental protection	30
1.5	Engine-Transmission Package - Lockout/Tagout and Unlocking Procedures	35
1.5.1	Lockout/tagout procedure	35
1.5.2	Unlocking procedure	37
2	Maintenance	39
2.1	Maintenance task reference table [QL1]	39
3	General Information	41
3.1	Engine side and cylinder designations	41
4	Design and Function	43
4.1	Frac Pack Overview	43
4.2	Sensors, actuators and injectors – Overview	45
5	Functional Description	49
5.1	Functional Description - Diesel Engine	49
5.1.1	Product summary for engine models 12 V 4000 T94 and T94L	49
5.1.2	Crankcase with oil pan	51
5.1.3	Gear train	53
5.1.4	Crank drive	54
5.1.5	Cylinder head with injector	56
5.1.6	Valve gear	58
5.1.7	Fuel system with common-rail injection	60
5.1.8	Charge-air and exhaust system with exhaust gas recirculation (EGR)	62
5.1.9	Lube oil system	65
5.1.10	Cooling system	67
5.1.11	Engine management and monitoring	69
6	Operating Instructions	73
6.1	General Information	73
6.1.1	Engine layout	73
6.2	Technical Data	74
6.2.1	Engine data 12V 4000 T94, T94L	74
6.2.2	Firing order	78
6.2.3	Main engine dimensions	79
6.3	Operation	80

6.3.1	Preparations for recommissioning the engine after extended out-of-service periods (>3 months)	80
6.3.2	Putting the engine into operation after scheduled out-of-service-period	81
6.3.3	Operational checks.....	82
6.3.4	Engine – Starting in manual mode.....	83
6.3.5	Engine – Stopping in manual mode.....	84
6.3.6	After stopping the engine.....	85
6.3.7	Plant – Cleaning.....	86
6.4	Troubleshooting.....	88
6.4.1	Troubleshooting.....	88
6.4.2	Fault messages on the SAM display (optional).....	91
6.4.3	Fault messages of Engine Control Unit ECU 9 for Series 4000	92
6.5	Task Description - Diesel Engine.....	151
6.5.1	Engine.....	151
6.5.1.1	Engine – Barring manually.....	151
6.5.1.2	Engine – Barring with starting system	152
6.5.1.3	Engine – Test run	153
6.5.2	Cylinder Liner.....	154
6.5.2.1	Endoscopically examining cylinder liner.....	154
6.5.2.2	Instructions and comments on endoscopic and visual examination of cylinder liners	156
6.5.3	Valve Drive.....	157
6.5.3.1	Valve gear – Lubrication	157
6.5.3.2	Valve clearance – Check and adjustment.....	158
6.5.3.3	Cylinder head cover – Removal and installation	161
6.5.4	Injection Pump / HP Pump	162
6.5.4.1	HP pump – Filling with engine oil.....	162
6.5.5	Injection Valve / Injector.....	163
6.5.5.1	Injector – Replacement	163
6.5.5.2	Injector – Removal and installation.....	164
6.5.6	Fuel System.....	171
6.5.6.1	Fuel system – Venting.....	171
6.5.7	Fuel Filter	172
6.5.7.1	Replacing fuel filter.....	172
6.5.7.2	Fuel prefilter – Draining.....	173
6.5.7.3	Auxiliary fuel filter – Replacement.....	175
6.5.8	Exhaust Turbocharger	176
6.5.8.1	Exhaust turbocharger – Turbine housing check.....	176
6.5.8.2	Exhaust turbocharger (LP) – Turbine housing check.....	177
6.5.8.3	Exhaust turbocharger (HP) – Turbine housing check.....	186
6.5.9	Charge-Air Cooling.....	193
6.5.9.1	Charge-air cooler – Leak check	193
6.5.10	Air Filter	194
6.5.10.1	Air filter – Replacement.....	194
6.5.10.2	Air filter element – Removal and installation (optional).....	195
6.5.11	Air Intake	196
6.5.11.1	Service indicator – Signal ring position check (optional)	196
6.5.11.2	Emergency air-shutoff flaps – Functional check with electric actuation.....	197
6.5.12	Exhaust Gas Recirculation	198

	6.5.12.1	Exhaust flaps – Overview	198
	6.5.12.2	Exhaust flaps – Checking operation of actuators	200
6.5.13		Lube Oil System, Lube Oil Circuit.....	201
	6.5.13.1	Engine oil – Level check with oil dipstick.....	201
	6.5.13.2	Engine oil – Change	202
	6.5.13.3	Engine oil – Sample extraction and analysis	204
6.5.14		Oil Filtration / Cooling.....	205
	6.5.14.1	Engine oil filter – Replacement.....	205
	6.5.14.2	Centrifugal oil filter – Cleaning and filter-sleeve replacement.....	207
6.5.15		Coolant Circuit, General, High-Temperature Circuit.....	209
	6.5.15.1	Engine coolant – Level check.....	209
	6.5.15.2	Engine coolant – Change.....	210
	6.5.15.3	Engine coolant – Draining.....	211
	6.5.15.4	Exhaust flaps – Draining coolant from servomotors.....	213
	6.5.15.5	Engine coolant – Filling	214
	6.5.15.6	Engine coolant pump pressure relief port – Check	217
	6.5.15.7	Engine coolant – Sample extraction and analysis.....	218
	6.5.15.8	Engine coolant filter – Replacement.....	219
6.5.16		Low-Temperature Circuit.....	220
	6.5.16.1	Charge-air coolant – Level check.....	220
	6.5.16.2	Charge-air coolant – Change	221
	6.5.16.3	Charge-air coolant – Draining.....	222
	6.5.16.4	Charge-air coolant – Filling	225
	6.5.16.5	Charge-air coolant pump – Relief bore check	228
6.5.17		Belt Drive.....	229
	6.5.17.1	Drive belt – Condition check.....	229
6.5.18		Battery-Charging Generator	230
	6.5.18.1	Checking battery-charging generator	230
	6.5.18.2	Battery-charging generator drive – Drive belt tension adjustment	231
	6.5.18.3	Battery-charging generator drive – Drive belt replacement.....	232
6.5.19		Fan Drive.....	234
	6.5.19.1	Fan drive (option) – Drive belt check / adjustment.....	234
	6.5.19.2	Fan drive (option) – Drive belt replacement.....	236
6.5.20		Engine Mounting / Support.....	237
	6.5.20.1	Engine mounting – Check	237
6.5.21		Wiring (General) for Engine/Gearbox/Unit.....	238
	6.5.21.1	Engine wiring – Check	238
6.5.22		Accessories for (Electronic) Engine Governor / Control System	239
	6.5.22.1	Resetting CDC parameter and entering IIG with DiaSys®.....	239
	6.5.22.2	Engine governor and connectors – Cleaning.....	240
	6.5.22.3	Engine governor plug connections – Check	241
6.6		Task Description - Gear Box.....	242
	6.6.1	Gearbox / Transmission.....	242
	6.6.1.1	Gear box – Checking securing screws for firm seating	242
7		Appendix A.....	243
	7.1	Conversion tables.....	243
	7.2	Tolerances and wear limits list – General information	248
	7.3	Tightening specifications for screws, nuts and bolts.....	249

	7.4	Abbreviations	260
	7.5	MTU contact persons/service partners.....	263
8		Appendix B.....	265
	8.1	Special Tools	265
	8.2	Index.....	267

1 Safety

1.1 General conditions

General

Safety notifications (DANGER, WARNING, and CAUTION) are used in this manual to alert the owner and/or operator to special instructions concerning a particular procedure that may be hazardous if performed incorrectly.

In addition to understanding the instructions in this publication, the user is also responsible for ensuring compliance with local safety requirements and regulations.

The safety information contained in this section relates to the engine-transmission package described in this publication and covers:

- Types of safety notifications used in this publication
- Prerequisites for working on an engine-transmission package
- Working on an engine-transmission package
- Unauthorized modifications, add-on devices and replacement parts

Types of Safety Notifications Used in this Publication

Safety notifications (DANGER, WARNING, and CAUTION) are used in this manual to alert all individuals to special instructions concerning a particular procedure that may pose a hazard if performed incorrectly. These safety notifications alone cannot eliminate the hazards that they signal. Strict compliance with these special instructions and common sense operation are critical accident prevention measures. All safety notifications found on the equipment must be observed. Ensure that safety-related labels are legible and not obstructed by dirt, grease or other equipment.



In the event of immediate danger.
Consequences: Death or serious injury

- Remedial action



In the event of potentially dangerous situations.
Consequences: Death or serious injury

- Remedial action



In the event of dangerous situations.
Consequences: Minor injury or material damage

- Remedial action



Meaning of NOTICE

- Notice is used throughout the publication to communicate a requirement, recommendation or fact relating to proper use, transport, installation, operation, or maintenance of the equipment but is not hazard related.



Manufacturers' recommendations

- All manufacturers' safety notifications are to be followed regardless of format.

The engine-transmission package manufacturer cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method or operating technique that is not specifically recommended by the engine-transmission package manufacturer is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that the product will not be damaged or made unsafe by the transport, installation, commissioning, operation, lubrication, maintenance, repair or storage procedures that you choose.

The information, specifications and illustrations in this manual are on the basis of information available at the time it was written. The information, specifications and illustrations can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information before you start any work on the engine-transmission package.

Prerequisites for working on an engine-transmission package

For the remainder of this publication, the term “working”, “work” or “all work” is used in reference to work done on the engine-transmission package. “Work” or “all work” is defined as transport, installation, commissioning, operation, maintenance, and repair.



NOTICE

Persons performing work on equipment

- Every person performing work on the equipment should read and become acquainted with all cautions and symbols before operating or repairing this product.



NOTICE

Permission for working on the equipment

- Always obtain permission from the person in charge before commencing work on the equipment.

Working on an engine-transmission package

Do not work on this engine-transmission package unless you have read and understand the instructions and safety notifications contained in this publication. Failure to follow the instructions or heed the safety notifications could result in injury or death. Contact an authorized sales or service outlet for a replacement manual.

Correct use

The engine-transmission package is intended exclusively for the application specified in the contract or defined at the time of delivery. Any other use is considered improper use. The engine-transmission package is to be used in accordance with the manufacturer's instructions as specified in this manual. Correct use includes, but is not limited to, observation of and compliance with the maintenance instructions and observation of maintenance and safety requirements. The engine-transmission package manufacturer will not be liable for any losses and/or damages caused by improper use or actions. The engine-transmission package manufacturer will not be liable for any losses and/or damages caused by improper use or actions resulting from personnel while under the influence of alcohol, medication or any other mind altering substances.

Unauthorized modifications and third-party add-on devices and replacement parts



NOTICE

Replacement safety labels

- Please contact an authorized service outlet to determine how to obtain replacement labels for parts that have safety labels.



NOTICE

Manufacturers' safety notifications

- All manufacturers' safety notifications are to be followed.
- Unauthorized modifications, third-party add-on devices and replacement parts may compromise the integrity of the equipment, and may present a safety risk.

Only genuine spare parts from the engine-transmission package manufacturer should be used to replace components or assemblies.

The engine-transmission package manufacturer will accept no liability or warranty claims for any damage caused by unauthorized modifications, third-party add-on devices and/or replacement parts. In the event that damage is caused by the use of non-genuine replacement parts, no liability or warranty claims through the engine-transmission package manufacturer will be accepted.

1.2 Personnel and organizational requirements

Publication notice

For the remainder of this publication, the term “work”, “working” or “all work” is used in reference to work done on the engine-transmission package. “Work”, “working” or “all work” is defined as one or more of the following actions: transport, installation, commissioning, operation, maintenance, repair and/or storage.

For the remainder of this publication, the term “certified personnel” refers to a certified technician who has successfully completed training at the manufacturer's certified training center and who has also been instructed on the engine-transmission package work by means of this publication, with special attention paid to the safety notifications. The manufacturer's factory certified technician training classes are available in various locations to meet worldwide needs.

For the remainder of this publication, the term “operator” or “third party user” is generally the person involved with the transport, installation, commissioning, operation, maintenance and/or repair that does not require him/her be an engine-transmission package manufacturer's certified technician, but who must read and have full understanding of this publication, especially the safety notifications, before proceeding with any operation.

For the remainder of this publication the term “owner” or “person responsible for the engine-transmission package operation” is the person who either owns the piece equipment or facility within which it resides. He/she generally schedules or directs the work being done on or with the engine-transmission package.

For the remainder of the publication, “third-party” or “add-on” generally refers to items not designed by or purchased from the engine-transmission package manufacturer; whereas, “genuine” or “factory” means it was designed by or purchased from the engine-transmission package manufacturer. Generally, the use of third-party add-ons and non-genuine replacement parts is the sole responsibility of the owner.

In general, operational checks can be performed by non-certified personnel following the published instructions. All other work on the engine-transmission package must be carried out by properly certified personnel. For the safety of the certified personnel and all others in the vicinity of work being done on the engine-transmission package, it is imperative that this publication is provided to all certified personnel involved in any or all work done on the engine-transmission package.



NOTICE

Manufacturers' recommendations

- All manufacturers' safety notifications are to be followed regardless of format.

Personnel and instructional requirements

This publication must be issued to all personnel involved in the transport, operation, maintenance, repair and/or storage of an engine-transmission package.

Responsibilities of the operating, maintenance and repair personnel must be specified by the “owner”.

This publication must be kept near the engine-transmission package and accessible to all personnel involved in operation, maintenance, repair and/or transportation.

The personnel must be instructed on engine-transmission package operation and repair by means of this publication, and in particular, the safety notifications must be explained and understood.

Owners are responsible for instructing personnel who only occasionally work on this engine-transmission package, prior to requesting the work be performed.



NOTICE

Additional safety standards

- Additional national safety standards exist. The manufacturer recommends owners, operators and certified technicians become familiar with national safety standards, where available.
- Operators should refer to the safety requirements for operators in this publication.

Additionally, all personnel working on the engine-transmission package must ensure that all safety notifications and informational messages are read and understood before any work is performed on the engine-transmission package.

- Strictly adhere to the maintenance and repair schedule.
- Never carry out the maintenance and/or the repair schedule with the engine and/or engine-transmission package running, unless specified in the procedure.
- Ensure persons not involved in the work on the engine-transmission package keep clear.
- Never attempt to rectify faults or carry out repairs without the necessary certification and required special tools. For clarification or assistance, contact the nearest authorized service outlet.
- Use proper, calibrated tools.
- Never work on the engine-transmission package, engine or other engine-transmission package components without following proper lifting requirements.
- Ensure nobody is standing in the danger zone prior to barring over the engine. After completing the work on the engine-transmission package, check that all protective devices/safety guards have been installed and that all tools and loose parts have been removed from the engine-transmission package.
- Observe special cleanliness when conducting maintenance and/or repair work on the engine-transmission package plant. After completion of maintenance and repair work, make sure that no loose objects are in/on the engine-transmission package.